PEOPLE COUNTER | PROJECT NOTE

Approach, observed limitation, and next steps

APPROACH

The project detects people with YOLOv8, tracks them across frames with Deep SORT, and uses OSNet appearance embeddings to help associate identities. A configured counting line and crossing confirmation logic generate IN and OUT events. Results are saved as an annotated video and an event CSV.

OBSERVED LIMITATIONS

In one observed case, a child was not counted as OUT after disappearing for a long interval while occluded behind a woman. Person IDs were also inconsistent across parts of the clip. Occlusion can break a track, and appearance-based re-identification may not recover the original identity confidently.

WHAT I WOULD DO WITH MORE TIME

Evaluate the full video with labeled crossings; tune the detector, tracker, and crossing thresholds; test stronger occlusion-aware tracking and long-term identity association; and measure missed/extra events and ID switches. Review crowded and partially hidden crossings separately before relying on the headcount.

WHAT THIS PROJECT DEMONSTRATES

People counting is a pipeline: detection, multi-object tracking, identity association, camera-specific crossing geometry, and event logic all affect the final count. Re-identification can help reconnect tracks but cannot guarantee identity continuity through long occlusions.

Privacy note The publicly shared project note intentionally contains no CCTV frame or identifiable bystander image.